MEASUREMENT & DIGITAL CONTROL

Week 2: Dynamic characteristics of measurement systems

VNU - UNIVERSITY *of* ENGINEERING & TECHNOLOGY

Agenda

1. **System Dynamics**
   Static vs. dynamic characteristics – ODE modeling
2. **First-order Systems**
   Zero-order & first-order systems – The time constant
3. **Second-Order Systems & Transient Response**
   Natural frequency & damping ratio – Step response - Rise time, overshoot, settling time
4. **Frequency Response**
   Bode plots – Bandwidth & phase lag
5. **Sensor selection**
   Types of sensors – Matching sensors to dynamic requirements

01

System Dynamics

Static vs. dynamic characteristics – ODE modeling

Static vs. Dynamic Characteristics

Static

Definition: Describe system behavior when the input is constant or slowly varying.

Purpose: Evaluate measurement correctness via:
- *Accuracy*
- *Precision / Repeatability*
- *Linearity*
- *Etc.*

Example: A thermometer measuring a stable room temperature.

Dynamic

Definition: Describe system behavior when the input is rapidly varying with time.

Purpose: Evaluate tracking ability via:
- *Time constant / Rise time / Settling time*
- *Overshoot*
- *Bandwidth*
- *Etc.*

Example: A thermometer measuring rapid temperature changes in a chemical reactor.

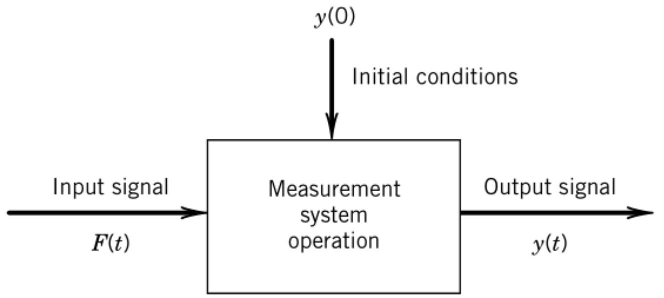
The General ODE of Measurement Systems

Differential equations are used to describe the dynamic behavior of linear time-invariant (LTI) systems

$$a_n \frac{d^n y}{dt^n} + a_{n-1} \frac{d^{n-1} y}{dt^{n-1}} + \cdots + a_1 \frac{dy}{dt} + a_0 y = F(t), \text{ where } F(t) = b_m \frac{d^m x}{dt^m} + \cdots + b_1 \frac{dx}{dt} + b_0 x \text{ with } m \leq n$$

Variable Definitions

> y: Output quantity (voltage, displacement, …)
> F: Input measurand (force, temperature, pressure, …)
> a_i, b_j: System coefficients (depend on physical parameters)
> n: Order of the system



💡 **Key insight:** *The order of this ODE determines the system's dynamic class*

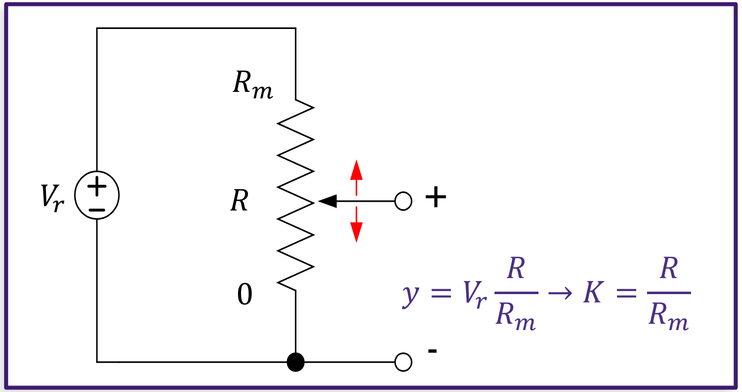
02

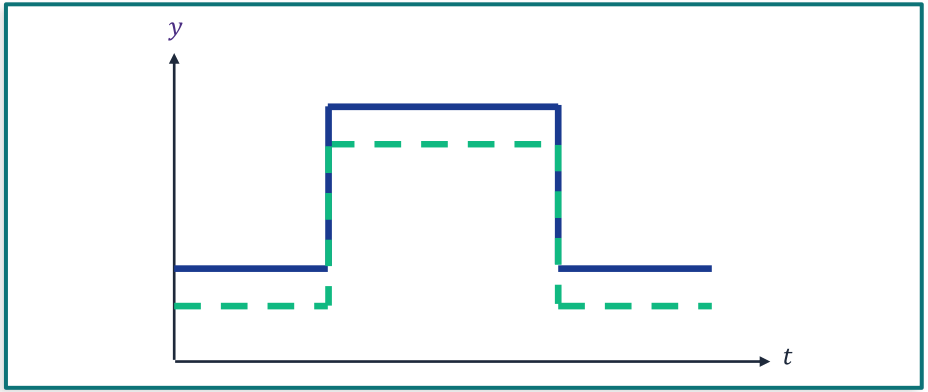
First-order Systems

Zero-order Systems - First-order Systems - The time constant

Zero-order Systems

$a_0 y = F(t) \implies y(t) = KF(t)$, where $K = \frac{1}{a_0}$ is the static sensitivity (steady gain) of the system.





Output is an instantaneous, scaled replica of the input | No time delay | Static sensitivity K fully characterizes behavior

First-Order Systems

$$a_1 \frac{dy}{dt} + a_0 y = F(t) \Longrightarrow \tau \frac{dy}{dt} + y = KF(t)$$

τ – Time constant

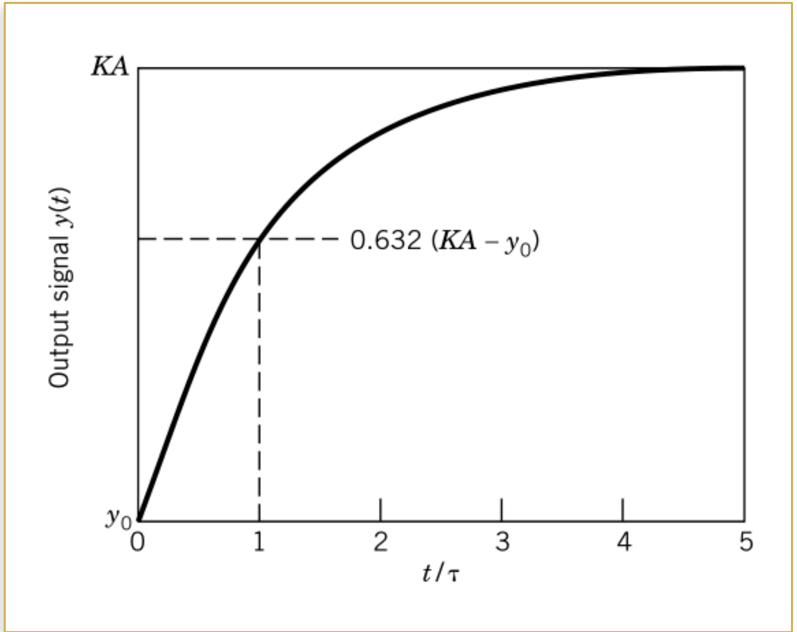
Controls how fast the system responds to a step change

K – Static Sensitivity

Ratio of output to input at steady state

Step response

- Input: $F(t) = Au(t)$
- Output: $y(t) = KA + (y_0 - KA)e^{-t/\tau}$
- $t = \tau$: 63.2% of final value



The Time Constant — Physical Meaning

τ

Physical meaning of τ

> The time it takes for the output to reach 63.2% of its final value after a step input.

> Determined by the ratio of the storage element (capacitance, thermal mass, inertia) to the dissipation element (resistance, convection coefficient, damping)

> Smaller τ means faster response (but may cost bandwidth, noise immunity)

τ in physical systems

System	τ Expression	Fast Response	Slow Response
RC Electrical Circuit	$\tau = R \cdot C$	Small R or C	Large R or C
Thermocouple (convection)	$\tau = m \cdot c_p / (h \cdot A)$	Small mass, large h	Large thermal mass
Mechanical (damper)	$\tau = m / b$	Light, high damping	Heavy, low damping

03

Second-Order Systems & Transient Response

Natural frequency - Damping ratio – Step response shapes - Performance metrics

Second-Order Systems — Governing Equation

$$a_2 \frac{d^2y}{dt^2} + a_1 \frac{dy}{dt} + a_0 y = F(t) \Longrightarrow \frac{1}{\omega_n^2}\frac{d^2y}{dt^2} + \frac{2\zeta}{\omega_n}\frac{dy}{dt} + y = KF(t)$$

ω_n Natural Frequency

- Frequency of free oscillation (rad/s)
- Higher $\omega_n \rightarrow$ faster system
- Determined by stiffness/inertia ratio

ζ Damping Ratio

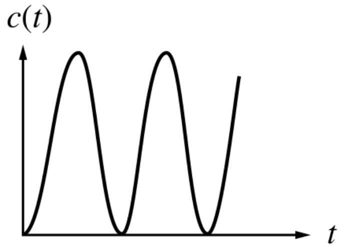
- Dimensionless
- Controls oscillation decay
  - $\zeta = 0$: undamped
  - $0 < \zeta < 1$: underdamped
  - $\zeta = 1$: critically damped
  - $\zeta > 1$: overdamped

K Static Sensitivity

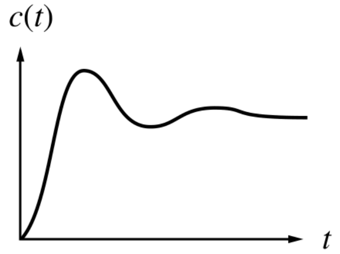
- Steady-state gain
- Same interpretation as in zero- and first-order systems

Step Response: Damping Cases

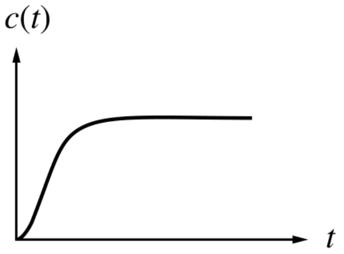
Undamped



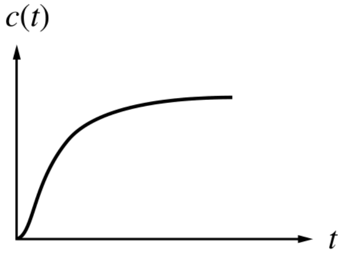
Underdamped



Critically damped



Overdamped



Transient Response Performance Metrics

Rise Time t_r

- Time to go from 10% to 90% of the final value.
- Indicates how quickly the sensor begins to respond.

$$t_r \approx (1 - 0.4\zeta)/f_n$$

Peak Time t_p

- Time to first reach the maximum (peak) output.
- Occurs at the first overshoot peak for underdamped systems.

$$t_p = \pi/\left(\omega_n\sqrt{1-\zeta^2}\right)$$

Overshoot Mp

- Percentage the output exceeds the final value
- Large overshoot may indicate instability

$$M_p = e^{-\zeta\pi/\sqrt{1-\zeta^2}} \times 100\%$$

Settling Time t_s

- Time for output to stay within ±2% (or ±5%) of the final value.
- Most critical spec for control applications.

$$t_s \approx 4/(\zeta\omega_n)$$

04

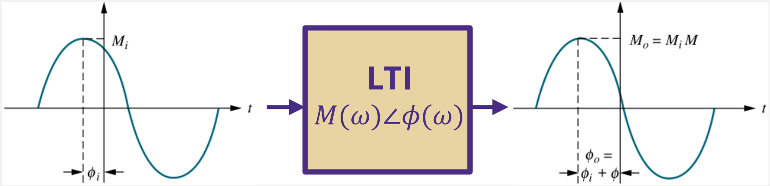
Frequency Response

Bode plots – Bandwidth - Phase lag

Frequency Response — Core Concept

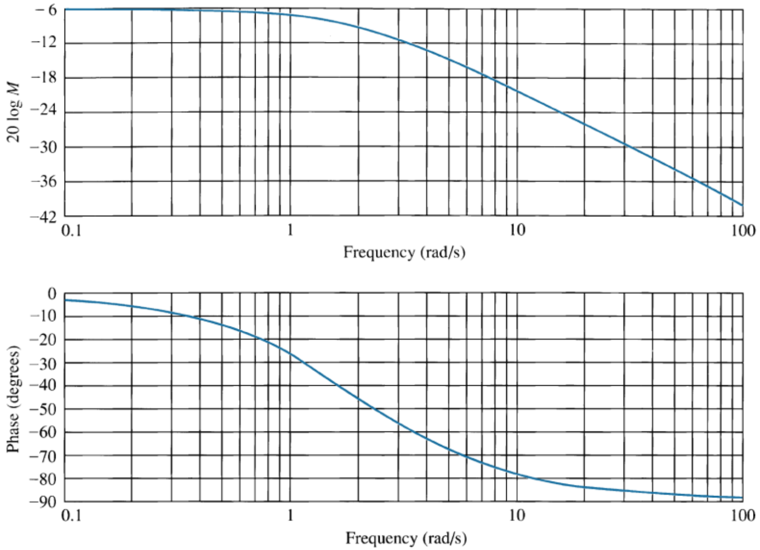
Transfer Function

- A linear system's steady-state response to a sinusoid is a sinusoid of the *same* frequency.
  - Differ in amplitude and phase



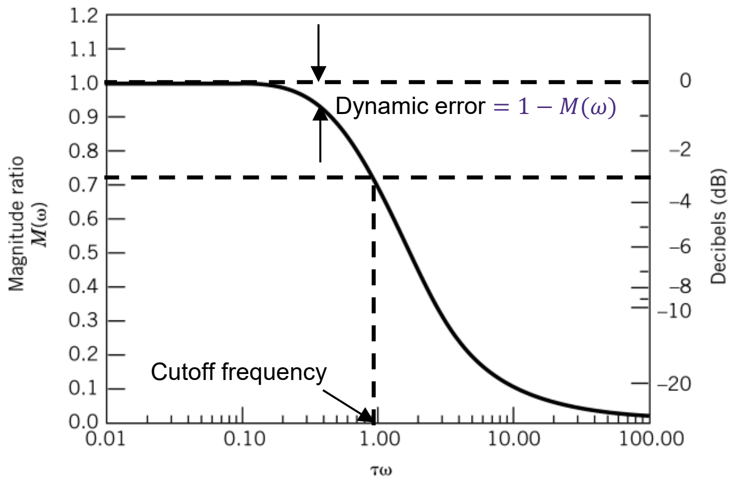
- Frequency response:
  - Magnitude response: $M(\omega) = M_o(\omega)\ / M_i(\omega)$
  - Phase response: $\phi(\omega) = \phi_o(\omega) - \phi_i(\omega)$
  - Both change with excitation frequency

Bode Diagram

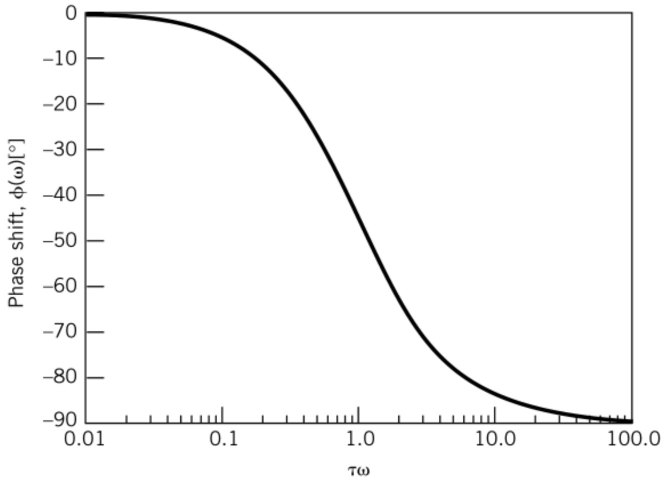


First-order System Frequency Response

Magnitude



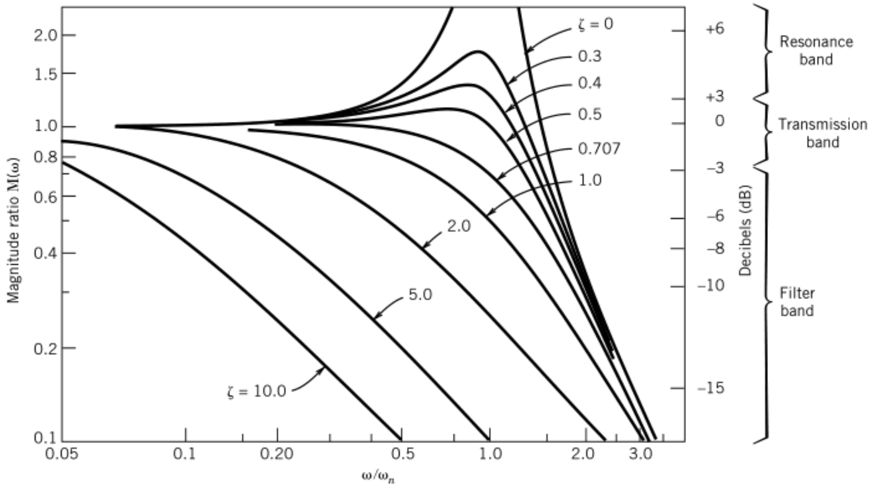
Phase



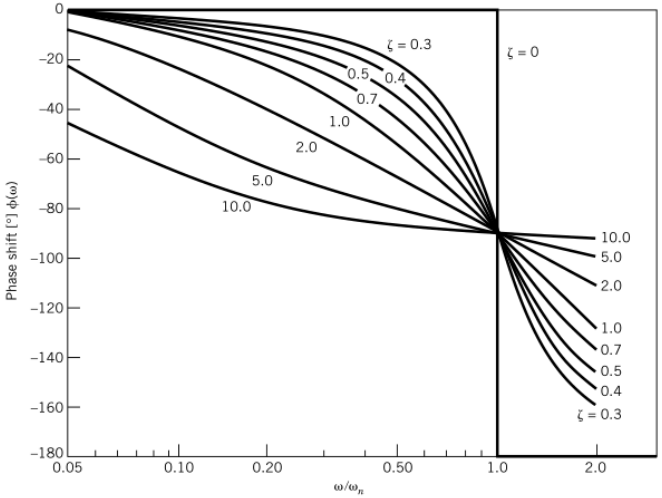
💡 **Key insight:** *Display low-pass filter behavior* $M(\omega) = \frac{1}{\sqrt{1+(\omega\tau)^2}}$

Second-order System Frequency Response

Magnitude



Phase



💡 **Key insight:** Also display low-pass filter behavior $M(\omega) = \frac{1}{\sqrt{[1-(\omega/\omega_n)^2]^2+[2\zeta\omega/\omega_n]^2}}$

Resonance in underdamped systems occurs at the resonance frequency $\omega_R = \omega_n\sqrt{1-2\zeta^2}$

Bandwidth & Phase Lag

Bandwidth (BW)

- The frequency range over which the system accurately tracks the input.
- Defined as the frequency at which the magnitude ratio drops to −3 dB (≈ 70.7% of DC gain)
  - ✓ 1st-order system: $BW = 1/(2\pi\tau)$
  - ✓ 2nd-order system: $BW \approx \omega_n$ (depends on ζ)

Phase Lag

- The time delay between input and output, expressed as an angle.
  - ✓ 1st-order system: $\varphi = -\arctan(\omega\tau)$
  - ✓ 2nd-order system: $\varphi = \arctan\left[\frac{2\zeta(\omega/\omega_n)}{1-(\omega/\omega_n)^2}\right]$
- Causes time errors in dynamic measurements.

Effect of System Parameters on Dynamic Behavior

Parameter	Increase Effect	Decrease Effect	Trade-off
τ (1st order)	Slower, lower BW	Faster, higher BW	Speed vs. noise immunity
ω_n (2nd order)	Higher BW, faster	Slower, lower BW	Speed vs. cost/complexity
ζ (2nd order)	Less overshoot, slower	More oscillation, faster peak	Stability vs. rise time

05

Sensor Selection

Types of sensors – Matching sensors to dynamic requirements

Sensor System Orders at a Glance

System Order	ODE Form	Key Parameters	Step Response	Freq. Response	Typical Sensors
Zero-Order (n = 0)	$q_0 = Kq_i$	K only	Instantaneous scaled copy	Flat (all freq.) No phase lag	Potentiometer Ideal strain gauge
First-Order (n = 1)	$\tau dq_0/dt + q_0 = Kq_i$	τ, K	Exponential rise to KA	−20 dB/decade rolloff above $1/\tau$	Thermocouple RC circuit Pressure transducer
Second-Order (n = 2)	$(1/\omega_n^2)\ddot{q}_0 + (2\zeta/\omega_n)\dot{q}_0 + q_0 = Kq_i$	ω_n, ζ, K	Damping-dependent overshoot/decay	Resonance peak ($\zeta<1$), −40 dB/dec rolloff	Accelerometer Piezoelectric Mass-spring sensors

Sensor Selection for Dynamic Applications

1 Define Signal

Estimate the frequency content of the measurand. Use FFT of historical data or engineering knowledge.

2 Set BW Target

Sensor bandwidth should be ≥ 5× the highest signal frequency of interest.

3 Check Phase Req.

Determine allowable phase lag. Critical in control loops - excess phase causes instability.

4 Evaluate Sensors

Compare manufacturer specs: −3 dB frequency, resonant frequency, damping ratio.

5 Verify & Validate

Bench-test with known frequency sweep. Compare to model predictions.

📐 Rule of Thumb: Sensor natural frequency ω_n should be at least 3–5× the highest frequency in the signal. For control, aim for 10×

Example

Scenario: *You need to measure vibration in a rotating shaft with dominant frequency components at 10, 50, 200 Hz*

Q1: What bandwidth does the sensor need?

BW ≥ 5 × 200 Hz = 1,000 Hz minimum.
Target BW: ≥ 1 kHz (aim for 2 kHz for margin).

Q2: If using a 2nd-order accelerometer, what ω_n is needed?

$\omega_n \geq 3 \times (2\pi \times 200) \approx 3{,}770$ rad/s $\rightarrow$ $\omega_n \geq 600$ Hz.
Select a sensor with $f_n \geq 1{,}000$ Hz to be safe.

Q3: What damping ratio is optimal?

$\zeta = 0.6$–0.7 gives the flattest magnitude ratio over the widest frequency range before roll-off.
This minimizes both overshoot AND bandwidth restriction.

Q4: A spec sheet shows $f_n = 800$ Hz, $\zeta = 0.5$. Is it adequate?

Borderline. $f_n = 800$ Hz gives −3 dB point ≈ 1.02 kHz — marginal for 200 Hz measurement.
Recommend a sensor with $f_n \geq 1.2$ kHz.

KEY TAKEAWAYS

1 **Static vs. Dynamic**

Always ask: does the measurand change faster than the system can respond? If yes, dynamic analysis is mandatory

2 **System Order Matters**

The ODE order dictates the response type: zero-order for steady-state, first/second-order for time-varying signals.

3 **Time Constant**

τ controls first-order speed: Smaller $\tau \rightarrow$ faster, but higher noise sensitivity.

4 **Damping Ratio**

For second-order sensors, $\zeta \approx 0.6–0.7$ maximizes bandwidth and minimizes overshoot

5 **Bandwidth & Phase Lag**

BW must exceed signal BW by at least 5×. Phase lag in dynamic measurements leads to timing errors

6 **Sensor Selection**

Sensor $\omega_n \geq 3–5\times$ signal frequency for measurement; 10× for closed-loop control.

Next Lecture: Signal conditioning